



Student Planting Handbook

YPL October Chapter
October 2026



1 Quick Reference Seasonal Planting Calendar

Plant	Spring	Summer	Autumn	Winter
Jacaranda	Good	Avoid	Best	Good
Lawn / Turf Grass	Best	Good	Fair	Avoid
Cineraria (Senaria)	Fair	Avoid	Best	Good
Hollyhock (Marshmallow Majaw)	Best	Good	Fair	Avoid
Red Wormwood	Not stated	Not stated	Not stated	Not stated
Antirrhinum / Snapdragon (Interhem)	Good	Avoid	Best	Good
Hibiscus / Roselle (Karkadeh)	Best	Good	Fair	Avoid
Pepper: Hot & Bell	Best	Good	Fair	Avoid
Papaver glaucum (Diddahan)	Fair	Avoid	Best	Good
Pansy	Fair	Avoid	Best	Good
Mint (Damask Peppermint)	Best	Good	Good	Fair
Sunflower	Best	Good	Fair	Avoid
Mimosa Pudica	Best	Good	Fair	Avoid
Sweet Pea	Fair	Avoid	Best	Good
Basil (Baladi)	Best	Good	Fair	Avoid
Chamomile	Fair	Avoid	Best	Good
Marigold (Qateefa)	Best	Good	Fair	Avoid
Rudbeckia / Coneflower	Fair	Avoid	Best	Good
Dimorphotheca	Fair	Avoid	Best	Good
Chrysanthemum (Achwan)	Best	Fair	Good	Fair
Alyssum (Lisum)	Good	Avoid	Best	Fair
Gazania	Best	Good	Good	Fair
Gypsophila	Best	Fair	Good	Fair

2 Trees

1. Jacaranda (Jacaranda)

Spring	Summer	Autumn	Winter
Good	Avoid	Best	Good
Quantity on hand	2 trees (50 cm nursery stock)		
Seed depth	N/A; planted as young trees, not seed		
Spacing	0.5 m radius planting hole, 60-85 cm deep		
Days to germinate	N/A (nursery stock)		
Watering	Weekly deep watering (until 7-10 cm of surrounding soil is moist) until established at 24 months old, then rely mostly on rainfall. Water the drip-line, never the trunk.		
Fertilizer & soil	Annual NPK 10-10-10, light on nitrogen (companion grass already supplies N). Salt tolerance: keep under 2.5 dS/m, absolute max 4 dS/m.		
Maturity	Established after 24 months; tolerant of -6°C once mature		
Winter hardy?	Yes. Tolerates down to -6°C once established; ideal for our winters.		
YouTube video	https://www.youtube.com/watch?v=5HsBKWu8Twc		

Description

Trees take long periods of time to grow from seeds, so we bought two young Jacaranda Trees instead. Therefore, this will only cover the care needed for these trees.

Growing instructions

- Placement:** Jacaranda trees require well-draining slightly sandy soil, not hard or extremely wet clay soils found in almost all of our school, but there are a few locations with soil in between clay and arid hard sand. It likes slightly salty soil, so reduction of salts wouldn't be a big issue for us here. Also, Jacaranda trees have significant surface roots which might disturb roads so we should keep it a distance away from any school road or mud path. Note that the tree is nevertheless tolerant to muddy loamy and muddy soil, but it is not preferred.
- Watering & Sun Exposure:** The Jacaranda Tree requires full sunlight exposure, at least six to eight hours of sunlight per day. It can tolerate freezing temperatures down to -6°C but generally prefers hot climates. Which is ideal for us as the temperature rarely if ever goes below 0°C in our school. The tree requires watering once a week until it is established (24 months old) then it should be less frequent to let it depend on rainfall. Do not water right at the trunk. Focus on the branches coming out of it. Water deeply until 3-4 inches (7-10 cm) of the soil around it feels moist, use your fingers to test the moisture but wash it well after.
- Pruning:** Cut some secondary branches of the tree so that only a main strong one remains to ensure a strong base. Do that by cutting the excess branches on the tip that is coming out of the main trunk with a garden scissor. Do not use excessive pruning or it will harm the tree and cause its shape might possibly be ruined.
- Fertilizer:** The Jacaranda requires annual fertilizer; however, too much nitrogen might harm the flowering and fertilizing grass below the tree already supplies it with nitrogen. Note that we are very likely to have grass or other plants around the tree so no excessive fertilizer. For the amount and type of fertilizer to use, an NPK (10-10-10) mixture is best and for the amount you consult the fertilizer label instructions. Salt Tolerance: Keep under 2.5 dS/m. The absolute maximum is about 4 dS/m.

3 Groundcover

1. Lawn / Turf Grass (Gazon)

Spring	Summer	Autumn	Winter
Best	Good	Fair	Avoid
Quantity on hand	1 kg		
Seed depth	Surface-broadcast, not buried		
Spacing	1 kg per 10 m ²		
Days to germinate	Establishes over several weeks with consistent moisture		
Watering	Frequent, light watering during germination/establishment (well matched to the new campus irrigation system); moderate once established.		
Fertilizer & soil	Nitrogen-rich lawn feed a few times through the growing season.		
Maturity	Full lawn coverage after one full growing season		
Winter hardy?	No. Goes semi-dormant and browns in cool weather; survives winter but won't fill in.		
YouTube video	https://youtu.be/FJYDXC9x8z4		

4 Herbs & Edibles

1. Hibiscus / Roselle (Karkadeh)

Spring	Summer	Autumn	Winter
Best	Good	Fair	Avoid
Quantity on hand	seed packets		
Seed depth	Shallow sow once soil is reliably warm		
Spacing	60-90 cm apart		
Days to germinate	Germinates readily in warm soil		
Watering	Moderate-to-regular; avoid waterlogged roots.		
Fertilizer & soil	Balanced fertilizer with potassium/phosphorus support for flower/calyx production.		
Maturity	Flowers/calyces (used for karkadeh tea) form in the warm season		
Winter hardy?	No. Frost-sensitive warm-season plant; will not survive a cold snap.		
YouTube video	https://youtu.be/7wLTHAYcCBI		

2. Pepper: Hot & Bell (Flefel Har / Flefel Roumy)

Spring	Summer	Autumn	Winter
Best	Good	Fair	Avoid
Quantity on hand	2-7 g		
Seed depth	1/4 in deep at transplant, once roots reach 3-6 cm		
Spacing	Standard pot spacing; final bed spacing 30-45 cm		
Days to germinate	Days to first white roots vary; check daily once soaking/wrapping begins		
Watering	Keep moist (check 3 in deep with a finger); do not overwater; causes root rot.		
Fertilizer & soil	Soil mix of 1/3 garden soil (not clay), 1/3 organic fertilizer (vermicompost/manure), 1/3 sand for drainage.		
Maturity	Fruit forms through the warm season; same care for hot and bell varieties		
Winter hardy?	No. Frost-sensitive; germination and growth both stall in cold soil.		
YouTube video	https://www.youtube.com/watch?v=EnlnaHMF7Y		

Growing instructions

- Germination:** Start by soaking the seeds in warm water overnight to give them a head start. After that, cover them in a damp water towel and lock them in a zip bag or container to trap humidity. Afterwards put the container or zip bag in a warm spot and check daily for it to grow white roots.
- Plantation:** Transfer the germinated seeds to the soil 1/4 inch deep after they reach a length of 3-6 cm. Pepper prefers well-draining with a mixture of 1/3 garden soil, not clay, 1/3 organic fertilizer, such as vermicompost or cow manure, and 1/3 sand to improve the drainage.
- Care:** Water the plants when dry to keep it moist, measure dryness with your finger 3 inches deep into the soil. Do not water excessively as it can cause root rot.

3. Mint (Damask Peppermint) (Naanaa)

Spring	Summer	Autumn	Winter
Best	Good	Good	Fair
Quantity on hand	7 units		
Seed depth	Transplant once 5-7 cm tall (50-70 days after germination)		
Spacing	18-24 in between rows, or 24 in unpotted		
Days to germinate	10-14 days		
Watering	Keep moist, water when the top 2-3 in of soil dries out.		
Fertilizer & soil	Peat moss improves drainage; avoid over-fertilizing; dulls the scent.		
Maturity	Fast-spreading perennial; usable for tea/harvest within one season		
Winter hardy?	Yes. Hardy perennial herb; may slow down but should survive our mild winters.		
YouTube video	https://www.youtube.com/watch?v=3VNv64iTRtQ		

Description

One of the highly invasive species. Spreads quickly and may take over other plant's nutrients and cause them to die. More ideal to plant within pots to limit their spread. However, the school has vast lands not covered with greenery where we can utilize the invasive nature of peppermint. Furthermore, it is a very versatile plant that can grow in almost any soil and circumstance, from full sun to partial shade to acidic soils to alkaline soils.

Growing instructions

- Germination:** Same method as Cineraria with a few tweaks. Close the container after gently tapping with a see-through plastic lid with air holes carved into it. Make sure to keep the soil moist and place it in a bright location. Usually sprouts in about 10-14 days.
- Plantation:** Transfer the seedlings from the sprouting pot after they grow to a decent size of more than 5-7 cm or after 50-70 days of germination to outside soil. If planting several plants in rows, allow for 18 to 24 inches between rows. Additionally, if not planted in a container, space it about 24 inches apart from other plants.
- Care:** Keep moist by watering the mint when at least 2-3 inches deep in the soil gets dry. Watch the overgrowth of peppermint and make sure to keep it in a container and/or keep it away from other plants grown nearby.
- Fertilization:** Peat moss is extremely beneficial as it improves the drainage of the soil. However, it is best to avoid over-fertilization as it can affect the scent of peppermint.

4. Sunflower (Dawar El Shams)

Spring	Summer	Autumn	Winter
Best	Good	Fair	Avoid
Quantity on hand	30 g		
Seed depth	Standard cup/pot sowing, lightly covered		
Spacing	Transplant once it outgrows the germination cup		
Days to germinate	1-2 weeks		
Watering	Keep moist; at least 3 in (7-10 cm) of soil wet at all times.		
Fertilizer & soil	Organic compost such as vermicompost, applied periodically.		
Maturity	Heads brown and are ready to harvest for seed later in the season		
Winter hardy?	No. A warm-season annual; frost will kill seedlings.		
YouTube video	https://www.youtube.com/watch?v=0FgNWD0upTc		

Growing instructions

- Germination:** Put a seed starter mix in a plastic cup so it almost comes up to the top. Then poke your finger in the soil to make a hole for your sunflower seed. Drop a sunflower seed into the hole. Cover it with a little soil and tap it gently. Spray or sprinkle some water on top. Put the cup somewhere sunny. Spray the soil with water whenever it feels dry. Your plant should pop up through the soil after about 1 or 2 weeks.
- Plantation:** Transfer the seedling to a pot or garden soil after it outgrows the germination cup. Water & keep moist. Make sure at least 3 inches (7-10 cm) of soil is wet.
- Care:** When the sunflower heads start to turn brown, cut them off and hang them somewhere dry and airy. When the seeds are dry, rub them together to loosen them and collect the seeds for future use.
- Fertilization:** Use organic compost such as vermicompost to fertilize the sunflowers from time to time.

5. Basil (Baladi) (Reihan Baladi)

Spring	Summer	Autumn	Winter
Best	Good	Fair	Avoid
Quantity on hand	15 g		
Seed depth	6-7 mm deep		
Spacing	Mulch 7.5-13 cm deep around plants		
Days to germinate	5-7 days; transplant once 2-3 true-leaf pairs form		
Watering	Keep soil moist; water freely during dry summer stretches.		
Fertilizer & soil	Sparing high-potassium fertilizer through the season.		
Maturity	Continuous harvest once established, with regular pruning		
Winter hardy?	No. Frost-sensitive warm-season herb.		
YouTube video	https://www.youtube.com/watch?v=eiaog6ctM1Q		

Growing instructions

- Sowing:** Sow no more than 6-7 mm deep, in February to mid-summer. Seeds should germinate within 5 to 7 days. Once they have developed 2 to 3 pairs of true leaves, transplant them.
- Mulch depth:** A 13- to 7.5-cm mulch of compost or ground-up leaves retains soil moisture and minimizes weeds around the plants.
- Soil moisture:** Make sure that the soil is moist.

4. **Hot locations:** Since it will be planted in hot areas, use mulch around the plants. The mulch will help hold in moisture and suppress weeds.
5. **Summer watering:** During the dry periods in summer, water the plants freely.
6. **First pruning:** After the seedlings have produced their first six leaves, prune above the second set.
7. **Ongoing pruning:** Every time a branch has six to eight leaves, repeat pruning the branches back to their first set of leaves.
8. **Fertilizer:** Fertilize sparingly throughout the season with a high-potassium fertilizer.

6. Chamomile (Al Kamomeel)

Spring	Summer	Autumn	Winter
Fair	Avoid	Best	Good
Quantity on hand	3 g		
Seed depth	Pressed onto the surface; needs light, do not bury		
Spacing	20-30 cm apart at transplant		
Days to germinate	1-2 weeks		
Watering	Mist regularly to keep evenly moist; keep well-draining soil.		
Fertilizer & soil	No heavy feeding needed; well-draining soil matters more.		
Maturity	Transplant outdoors once 2.5-5 cm tall, after 1-2 weeks hardening off		
Winter hardy?	Yes. A cool-season herb well suited to our winters.		
YouTube video	https://www.youtube.com/watch?v=ibXazvct4KI&t=253s		

Growing instructions

1. **Prepare trays:** Fill up seed-starting trays or pots with a pre-moistened seed-starting mix, planted in September.
2. **Sowing:** Leave the seeds pressed into the surface, to provide them light for germination.
3. **Germination:** Move the seeds under grow lights and mist the soil regularly to keep it evenly moist. Seeds should sprout in about 1 to 2 weeks.
4. **Hardening:** Once the seedlings are 2.5 to 5 cm tall, harden your plants off for 1 to 2 weeks, and then transplant them outdoors into well-draining soil.
5. **Transplanting:** When transplanting, bury the plants' root balls at the same depth they were growing in their seedling trays or pots. Plant them 20-30 cm apart.
6. **Indoor growth:** Don't let the indoor-started chamomile seedlings grow too long.

5 Flowers

1. Cineraria (Senaria)

Spring	Summer	Autumn	Winter
Fair	Avoid	Best	Good
Quantity on hand	seed packets		
Seed depth	Surface-sown, covered with 3x seed thickness of mix		
Spacing	3 in apart at transplant		
Days to germinate	8-10 days; transplant 3 weeks after sprouting		
Watering	Water when the top 1 in (3 cm) of soil is dry. Never wet the leaves; fungal risk.		
Fertilizer & soil	Peat moss + organic fertilizer. Salt tolerance up to 11.5 dS/m (growth slows near that limit).		
Maturity	Blooms through the cool season once transplanted		
Winter hardy?	Yes. A cool-season bloomer; thrives in our mild winters.		
YouTube video	https://www.youtube.com/watch?v=NK_eykACiRc		

Growing instructions

- Germination:** Prepare a seed starting mix in a pot with drainage holes. If there aren't any then feel free to drill 4-6 small holes around the bottom of the pot. The seed starting mix should have 70% Peat Moss and 30% Vermicompost and additionally one spoon of fungicide powder. After preparing the mix pack it inside the pot firmly and eliminate any gaps in the soil. Then moisturize the soil with water well. Sprinkle the seeds evenly on top of the soil to ensure maximum germination efficiency. After spreading the seeds cover them with three times the thickness of the same soil mix then tap it gently. After covering sprinkle more water on the mix but little by little as to not displace the seeds. Keep the pot in partial sun for a 10-15 day period and keep it moist by watering it as it dries. The seeds usually germinate and sprout in 8-10 days. As the seedlings grow gradually transfer them to areas with more sunlight. The ideal time to transplant the seeds is after 3 weeks of sprouting.
- Plantation:** Transplant the germinated seeds from the pots with a gardening tool or any stick without disturbing the roots of the plant and then transfer it to the permanent soil. The ideal soil for the transplanted cineraria plants is composed of 30% Compost, 50% Garden soil, 20% Sand/Perlite. Level the transplanted seedlings with the soil surface and make sure the base is above the soil surface. For maximum growth, keep plants away from each other by 3 inches each.
- Watering & Sun Exposure:** Cineraria is different from the Jacaranda trees, almost the quite opposite. It requires partial sunlight exposure, safe maximum of 4-6 hours of sunlight, or shade. Lots of shade can undersupply the plant with nutrients. It requires constant moisture. Water the drainage or soil of the plant when the first inch, about 3 cm, feels dry with your finger. Do not wet the leaves nor turn the soil soggy as it can cause fungal infections.
- Fertilization:** Use peat moss and organic fertilizers to provide the rich and well drained soil for the plant. The amount depends on the instructions on the back of the packages and soil test readings of the spot. The soil should have good drainage, water does not get clogged. Salt Tolerance: Up to 11.5 dS/m. However high levels near the limit hinder plant growth.

2. Hollyhock (Marshmallow Majaw)

Spring	Summer	Autumn	Winter
Best	Good	Fair	Avoid
Quantity on hand	seed packets		
Seed depth	1/8 in deep		
Spacing	18-24 in (45-61 cm) apart		
Days to germinate	10-14 days		
Watering	Constant moisture; cut spent stalks to the ground to extend the plant's life.		
Fertilizer & soil	Peat moss + organic fertilizer. Salt tolerance: safe to 6 dS/m, tolerates up to 12 dS/m.		
Maturity	Biennial/short-lived perennial; blooms in its second season once established		
Winter hardy?	Yes. Established plants tolerate mild winter; seedlings need wind shelter.		
YouTube video	https://www.youtube.com/watch?v=9S0rEh2U7W0		

Description

Hollyhocks prefers a warm, sunny location sheltered from the wind, because of that we must plant it in areas that are more sheltered from wind.

Growing instructions

1. **Germination:** Very similar to Cineraria, refer to the previous method and ensure moisture and cover the seeds with triple the seed width of soil. Use 70% Peat moss 30% Vermicompost seed starting mix. Hollyhocks usually take from 10 to 14 days to sprout. Requires light shade or partial sun, the sunlight requirement increases gradually with size.
2. **Plantation:** Space the plants 18 to 24 inches (45-61 cm) apart and plant each seedling 1/8 inch deep in evenly prepared soil. A good soil composition would be 30% Compost, 50% Garden soil, 20% Sand/Perlite for its good drainage and nutrients.
3. **Care:** Needs constantly moist soil, water when the soil dries out. Use the previous methods of 3-1 dry inches of soil with your finger to check the moisture. When the flowers begin to fade, cut the stalks to the ground to increase the plant's lifespan.
4. **Fertilization:** Use peat moss and organic fertilizers to provide the rich and well-drained soil for the plant. The amount depends on the instructions on the back of the packages and soil test readings of the spot. The soil should have good drainage, water does not get clogged. Salt Tolerance: Safe limit 6 dS/m. Can tolerate up to 12 dS/m.

3. Antirrhinum / Snapdragon (Interhem)

Spring	Summer	Autumn	Winter
Good	Avoid	Best	Good
Quantity on hand	seed packets		
Seed depth	Surface-sown; needs light to germinate		
Spacing	15-30 cm apart		
Days to germinate	1-2 weeks at cool-mild soil temperatures		
Watering	Moderate, consistent watering; avoid drying out completely, but don't waterlog.		
Fertilizer & soil	Balanced fertilizer at planting, light monthly feed during bloom.		
Maturity	Blooms autumn through spring; bolts and stops flowering in real heat		
Winter hardy?	Yes. One of the more cold-tolerant bedding flowers; light frost doesn't bother it.		
YouTube video	https://youtu.be/9D3H-HpQYDE		

4. Red Wormwood (Red Wormwood)

Spring	Summer	Autumn	Winter
Not stated	Not stated	Not stated	Not stated
Quantity on hand	Not stated		
Seed depth	1/2 in deep		
Spacing	18 in (50 cm) apart; mature spread 1-2 ft		
Days to germinate	90-120 days from seedling to flowering maturity		
Watering	Keep moist during germination; water weekly after transplant and adjust using a soil-moisture check.		
Fertilizer & soil	Use only small amounts of organic fertilizer such as vermicompost. Peat moss may be added for drainage.		
Maturity	Flowering plant after approximately 90-120 days		
Winter hardy?	Winter hardiness is not stated.		
YouTube video	https://www.youtube.com/watch?v=92lrnPPkOcs		

Description

This is a low effort plant that does not thrive in highly nutritious or heavy soils. Instead, it grows well in poor to moderately fertile soils. However, it still requires well drainage with a small to moderate amount of moisture to grow.

Growing instructions

1. **Germination:** Plant the seeds 1/2 inch deep in moderately fertile soil (6.0 to 7.5 pH) and keep them moist, water in small amounts when at least 3 inches of the soil are dry, in a sunny location until they sprout. Seedlings will take 90 to 120 days to mature into flowering plants.
2. **Plantation:** Transfer the seedlings with a farming tool or stick without interfering with the seeds and put in well-drained soil of moderate fertility, remember that highly fertile soil hinders the growths of red wormwood. Keep plants 18 inches (50 cm) away from each other to allow them to reach the maximum spread of 1 to 2 feet.
3. **Care:** Water weekly and/or based on the moisture of the plant's soil. Same technique used to measure the moisture of other plants can also be used here.

4. **Pruning:** In early to late spring cut dead or damaged parts of the plant. Additionally, cut branches in overcrowded areas to increase sunlight reach and support proper growth of the plant.
5. **Fertilization:** Small amounts of organic fertilizer can be added to support the plant such as vermicompost. Peat Moss can be added to improve drainage as well.

5. Papaver glaucum: Red Wild Poppy (Diddahan)

Spring	Summer	Autumn	Winter
Fair	Avoid	Best	Good
Quantity on hand	4 x 10 seeds		
Seed depth	Surface-sown or barely covered		
Spacing	Broadcast-style; naturalizes in drifts		
Days to germinate	Low-input annual; no special germination pampering needed		
Watering	Low-to-moderate; a desert/steppe wildflower; avoid overwatering once seedlings are up.		
Fertilizer & soil	Minimal; tolerates poor soil well.		
Maturity	Blooms in the cool season, naturalizing over successive years if allowed to self-seed		
Winter hardy?	Yes. A native cool-season wildflower; exactly the hardy, regionally native plant Moghny's notes recommend.		
YouTube video	https://youtu.be/GAmvTvc-RnQ		

6. Pansy (Pansy)

Spring	Summer	Autumn	Winter
Fair	Avoid	Best	Good
Quantity on hand	seed packets		
Seed depth	Root ball level with soil surface		
Spacing	15-25 cm apart		
Days to germinate	Sown late summer/early autumn as temperatures cool		
Watering	Keep consistently moist, especially in containers or dry spells.		
Fertilizer & soil	Organic fertilizer (compost/well-rotted manure); optional liquid feed every 2-4 weeks.		
Maturity	Classic Egyptian winter bedding flower; blooms through the cool months		
Winter hardy?	Yes. A classic Egyptian winter bedding flower; plant it exactly for the cool months.		
YouTube video	https://www.youtube.com/watch?v=LDlnZqqtYyY		

Growing instructions

1. **Planting time and site:** Plant pansies in late summer or early fall, when temperatures begin to cool down. Plant in an area with well-draining slightly-acidic-to-neutral soil with full sun with some afternoon shade.
2. **Soil preparation:** Prepare the soil by loosening it to a depth of 20-30 cm and incorporating organic matter.
3. **Spacing:** Space the plants 15-25 cm apart.
4. **Planting hole:** Transplant into a hole slightly larger than the root ball of the plant.

5. **Planting depth:** Make the top of the root ball level with the soil surface. Press the soil down gently but firmly.
6. **Watering:** Keep soil consistently moist. Water regularly, especially during dry spells or when grown in containers.
7. **Organic fertilizer:** Use an organic fertilizer, such as compost or well-rotted manure.
8. **Optional liquid feed:** To promote continuous blooming, apply a liquid fertilizer every 2-4 weeks during the growing season.
9. **Deadheading:** Remove spent blooms regularly.
10. **Mulch:** Apply a layer of mulch around the base of the plants. Use organic mulch at a depth of 5-7 cm.

7. Mimosa Pudica (Touch-me-not) (Al Mostahia)

Spring	Summer	Autumn	Winter
Best	Good	Fair	Avoid
Quantity on hand	3 g		
Seed depth	0.3-1 cm deep		
Spacing	Pot individually; a creeping/trailing plant		
Days to germinate	1 week		
Watering	Water when the topsoil dries; keep moist, never waterlogged. Reduce watering in winter.		
Fertilizer & soil	No special feeding; focus on well-draining, loamy soil instead.		
Maturity	Hardens off after 3-4 sets of true leaves, then safe to move outdoors		
Winter hardy?	No. Sensitive to cold; water even less in winter if kept as an established plant.		
YouTube video	https://www.youtube.com/watch?v=Jhfo1LtwU7Y		

Growing instructions

1. **Seed preparation:** Scratch the seed's coat with a knife, and soak the seeds overnight in warm water. This can be done in late winter or early spring, mainly from March to May.
2. **Sowing:** Sow the seeds in small pots filled with moistened potting mix, around 0.3-1 cm deep. It has to be well-draining, loamy soil. Peat moss can preferably be used.
3. **Germination location:** Set the pot in a bright warm location. Germination usually takes only about one week.
4. **Watering:** Water typically when the top of the soil dries, keeping the soil moist but not waterlogged. Water less in winter.
5. **Hardening and transplanting:** After developing 3-4 sets of true leaves, the plant should be hardened off, by making it spend progressively longer periods of time outdoors in spots protected from intense sun and wind. Afterwards they can be transplanted safely outdoors.
6. **Pruning and training:** Prune regularly to keep it full and bushy rather than leggy. Since Mimosa pudica is a creeping plant, trim off the trailing stem or train them to climb a trellis to keep the plant looking neat.

8. Sweet Pea (Baslet El Zohour)

Spring	Summer	Autumn	Winter
Fair	Avoid	Best	Good
Quantity on hand	6 g		
Seed depth	1.25 cm deep, 2 seeds per pot		
Spacing	20 cm apart in rows, near 1.8 m fencing		
Days to germinate	Speeds up with a 24 hr pre-soak and a plastic dome cover		
Watering	Consistent moisture; drip irrigation/soaker hoses from planting onward. Water deeply 2-3x/week.		
Fertilizer & soil	Work compost into the soil at planting.		
Maturity	Vines climb rapidly once trellised; pinch central tip at 10-15 cm tall		
Winter hardy?	Yes. A classic autumn-sown climber for our winters; sow right at the start of the season.		
YouTube video	https://www.youtube.com/watch?v=Al-0Ae8AGKI		

Growing instructions

1. **Sowing period:** Sow sweet peas in September/October to December. Soak your seeds for 24 hours before sowing. This softens the seed coat and speeds the sprouting process by a few days.
2. **Prepare the pots:** While the seeds are soaking, fill your planting pots with soil.
3. **Plant the seeds:** Plant two seeds per pot, poking them 1.25 cm into the soil with your finger.
4. **Humidity:** Cover pots with a plastic dome lid to increase humidity and speed up germination.
5. **Pinching:** Once plants are 10 to 15 cm tall, pinch out the central growing tip, just above a leaf joint, leaving just 2 or 3 leaf nodes.
6. **Soil improvement:** In addition to the standard soil preparation, use compost.
7. **Site and support:** Vines grow rapidly and require a spot that gets full sun, as well as a structure to climb. Plant near 1.8-meter-high metal fencing.
8. **Planting out:** Around the last spring frost, plant out seedlings in rows, roughly 20 cm apart down the row. Keep them tied to the metal fencing to produce straight stems. Tying is a priority due to fast growth.
9. **Watering:** Sweet peas need consistent moisture. Drip irrigation or soaker hoses, use as soon as planted. Water deeply 2 to 3 times a week.

9. Marigold (Qateefa) (Qateefa)

Spring	Summer	Autumn	Winter
Best	Good	Fair	Avoid
Quantity on hand	2 g		
Seed depth	2-3 cm deep		
Spacing	Thin to 20-30 cm apart		
Days to germinate	Direct-sown, spring to midsummer (around April)		
Watering	Let soil mostly dry, then water well at the base (never overhead); more in high heat.		
Fertilizer & soil	Only if soil is nutrient-poor (slow-release feed in the planting hole); do NOT fertilize during active growth.		
Maturity	Continuous bloom with regular deadheading		
Winter hardy?	No. Can limp through a mild winter but growth/flowering both stall.		
YouTube video	https://www.youtube.com/watch?v=KecXc5rGVV4		

Growing instructions

1. **Prepare poor soil:** If soil is nutrient-starved, add a slow-release high-potassium fertilizer in the planting hole.
2. **Direct sowing:** Moisten the soil, then sow seeds 2-3 cm apart and no more than 2-3 cm deep. Sow directly into the ground in spring to mid-summer, around April.
3. **Thinning:** While still small, thin the seedlings. The plants should be 20-30 cm apart.
4. **Pinching:** When the marigolds grow and establish themselves, pinch off the tops of the plants.
5. **Deadheading:** Regularly remove dying blooms.
6. **Watering:** Allow the soil to mostly dry, then water well and repeat the process. Water more in high heat.
7. **Watering position:** Do not water marigolds from overhead. Water at the base of the plant.
8. **Fertilizer during growth:** Do not fertilize marigolds during growth.
9. **Mulch:** Preferably use mulch between plants to suppress weeds and keep the soil moist, especially when plants are young.

10. Rudbeckia / Coneflower (Rudbeckia)

Spring	Summer	Autumn	Winter
Fair	Avoid	Best	Good
Quantity on hand	700 seeds		
Seed depth	0.6 cm loose soil cover		
Spacing	Full sun spot, 4-5 hrs, with shade breaks		
Days to germinate	10-20 days		
Watering	Deep, frequent watering until established, then minimal; mulch helps retain moisture.		
Fertilizer & soil	Fertilize once in autumn only; avoid too much nitrogen.		
Maturity	Hardy perennial once established; a strong long-term winter-tolerant choice		
Winter hardy?	Yes. A hardy perennial once established; a strong long-term winter-tolerant choice.		
YouTube video	https://www.youtube.com/watch?v=EfZiGqfXKeU		

Growing instructions

1. **Sowing time:** Sow in September/October.
2. **Bed preparation:** Use organic matter and get rid of weed.
3. **Sowing:** Evenly sow seeds and cover the seed with 0.6 cm loose soil and gently press in soil.
4. **Germination care:** Keep soil evenly moist until seedlings emerge in 10-20 days.
5. **Watering:** Water deep and often until the coneflowers are established, then only a minimum amount of water is needed. Mulch around the base will aid in this.
6. **Fertilizer:** Fertilize once in fall, and avoid too much nitrogen.
7. **Sunlight:** Plant in an area with 4-5 hours of full sun with shade breaks in the morning/afternoon.

11. Dimorphotheca (African Daisy) (Dimorphotheca)

Spring	Summer	Autumn	Winter
Fair	Avoid	Best	Good
Quantity on hand	1 g		
Seed depth	Surface-sown, very lightly covered		
Spacing	At least 30 cm apart		
Days to germinate	Needs light to germinate; keep moist until sprouted		
Watering	Keep evenly moist, about 1 in of water per week.		
Fertilizer & soil	Rich, well-draining, slightly acidic soil; balanced fertilizer monthly plus compost.		
Maturity	Hardened off 2 full weeks before outdoor planting; pinch back at planting for bushier growth		
Winter hardy?	Yes. Cool-season annual, well suited to autumn sowing for winter/spring bloom.		
YouTube video	https://www.youtube.com/watch?v=RMV_ayyx5uU		

Growing instructions

1. **Sowing:** In September/October, sow the seeds directly on the surface of a pot filled with ordinary potting mix. Very lightly cover the seeds, light is necessary for the seeds to germinate.
2. **Germination moisture:** Keep the seeds moist until they germinate.
3. **Seedling light:** Grow the seedlings in bright indirect light until outdoor planting time.
4. **Hardening and pinching:** Plants started indoors should be hardened off for two full weeks before outdoor planting. Pinch back the seedlings immediately after planting to encourage bushier growth.
5. **Spacing:** Plant them at least 30 cm apart.
6. **Water, soil, and fertilizer:** Keep soil evenly moist, with 1 inch of water per week. They require rich well-draining soil, preferably slightly acidic. Balanced fertilizer monthly throughout the growing season along with mixing compost into the soil would be ideal.

12. Chrysanthemum (Achwan) (Achwan)

Spring	Summer	Autumn	Winter
Best	Fair	Good	Fair
Quantity on hand	2 g		
Seed depth	Standard sow, February-April		
Spacing	45-90 in apart depending on mature size		
Days to germinate	Sown February-April		
Watering	Soak the soil deeply; keep evenly moist but avoid wetting the leaves.		
Fertilizer & soil	Balanced 10-10-10 monthly; stop once buds appear, then switch to high-nitrogen after buds set.		
Maturity	Mulch after ground has cooled hard; new green shoots signal spring regrowth		
Winter hardy?	Yes. Mulch protects it through frost/cold snaps; established plants carry through winter reasonably well.		
YouTube video	https://www.youtube.com/watch?v=nfVnWF5nXMY&t=190s		

Growing instructions

1. **Sowing and spacing:** Sow seeds in February to April. Set them 45 to 90 inches apart, depending on their expected size at maturity.
2. **Watering:** Water and keep it evenly moist. Keep the chrysanthemums moist. Soak the soil deeply when watering. Avoid watering the leaves.
3. **Monthly fertilizer:** Apply a balanced fertilizer (10-10-10) monthly.
4. **Mulch:** Spread mulch around the plant.
5. **Bud stage:** Stop fertilizing when flower buds emerge.
6. **Large flowers:** For big flowers, pinch off all buds except the largest and strongest on each stem. Remove secondary buds farther down the stem as they develop, too.
7. **After bud set:** Switch to a high-nitrogen fertilizer after the buds have set.
8. **Winter mulch:** After the ground has frozen, spread 10 to 15 inches of mulch around the plant.
9. **Spring cleanup:** Cut off the dead stems and leaves when you see the first green shoots in spring.

13. Alyssum (Lisum)

Spring	Summer	Autumn	Winter
Good	Avoid	Best	Fair
Quantity on hand	1 g		
Seed depth	2 mm deep		
Spacing	20 cm apart (or scatter-sown)		
Days to germinate	7-14 days at 18-22°C soil temperature		
Watering	Keep moist but never wet or dry; water deeply in early morning or late afternoon.		
Fertilizer & soil	Only needed in poor soil; slow-release high-potassium feed at 5-10 cm tall, or liquid feed while flowering.		
Maturity	Flowers in 60 days from sowing		
Winter hardy?	Yes. Reasonably cold-tolerant; young seedlings need protection until established.		
YouTube video	https://www.youtube.com/watch?v=47Qy_rK8Spl&t=265s		

Growing instructions

1. **Direct sowing:** Sow seeds directly in the garden 2 mm deep and 20 cm apart.
2. **Moisture:** Keep soil moist but never wet or dry.
3. **Germination:** Seeds should germinate in around 7-14 days at a soil temperature of 18-22°C.
4. **Seedling protection:** Young seedlings will need protection from pests, pets and weather until they are established.
5. **Scatter method:** Alyssum seeds can also be sown in the less formal scatter seed method. Simply roughen the soil, scatter seeds evenly over the surface, then smooth the soil over lightly to cover the seeds.
6. **Watering:** Water when the soil is dry about 5 cm below the surface. Water deeply in the early morning or late afternoon.
7. **Fertilizer:** Apply slow release high-potassium fertilizer when seedlings are 5-10 cm tall. Or apply liquid fertilizer while plants are fruiting or flowering. Only do in poor soil, otherwise none needed.
8. **Optional thinning:** Thin seedlings when they are large enough to handle. Pull out any weak or small seedlings so plants are spaced about 20 cm apart.
9. **Flowering:** Alyssum plants should flower in approximately 60 days.
10. **Deadheading:** Deadhead alyssum flowers regularly.

14. Gazania (Gazania)

Spring	Summer	Autumn	Winter
Best	Good	Good	Fair
Quantity on hand	0.75 g		
Seed depth	Not documented by the team yet; general reference only		
Spacing	20-30 cm apart, spreads as groundcover		
Days to germinate	Not documented by the team yet		
Watering	Low water needs once established; good for sunnier, less-irrigated beds.		
Fertilizer & soil	Light feeding only; heavy fertilizer pushes leaves over flowers.		
Maturity	Spreads as a low groundcover once established		
Winter hardy?	Yes. Tolerates mild winters well; dislikes cold, waterlogged soil more than cold air itself.		
YouTube video	https://youtu.be/eMp7jxlVMr4		

15. Gypsophila (Baby's Breath) (Gypsophila)

Spring	Summer	Autumn	Winter
Best	Fair	Good	Fair
Quantity on hand	0.75 g		
Seed depth	Shallow sow; not documented by the team yet		
Spacing	30-45 cm apart		
Days to germinate	Not documented by the team yet		
Watering	Low-to-moderate; drought tolerant once established. Avoid overwatering; causes root rot.		
Fertilizer & soil	Minimal; rich soil produces leggy, floppy growth.		
Maturity	Airy filler growth habit, good for cut-flower demonstrations		
Winter hardy?	Yes. Moderately cold-tolerant; dislikes acidic, waterlogged soil more than cold itself.		
YouTube video	https://youtu.be/6WhpDuFD43M		

6 General Reference

Common soil & seed-starting mixes used across this handbook

Seed-starting mix	70% peat moss / 30% vermicompost + 1 spoon fungicide powder (Cineraria-style method, reused for several species)
Transplant soil	30% compost / 50% garden soil / 20% sand or perlite
Pepper soil mix	1/3 garden soil (not clay) / 1/3 organic fertilizer / 1/3 sand
Jacaranda fertilizer	NPK 10-10-10 annually, light on nitrogen
General salt-tolerance note	Most flower species in this handbook tolerate 6-12 dS/m; Jacaranda is the most sensitive at 2.5-4 dS/m max

YouTube video note

For our Muslim members of the YPL, some of these videos might contain music. Look for ones not containing music or remove the music from these ones. The YPL might, when there is time, provide you with a drive link with fully halal videos. However, in the meantime, you have been warned and the sin of listening to music from the videos below is completely on you. *Allahumma ballaght, Allahumma fashhad.*

Glossary

Hardening off	Gradually exposing indoor/potted seedlings to outdoor sun, wind, and temperature over 1-2 weeks before permanent transplant
Damping-off	A fungal seedling disease prevented by fungicide powder in the seed-starting mix and by avoiding overly wet, stagnant soil
Dead-heading	Removing spent flowers to encourage further blooming
Winter hardy	Able to survive our mild 6th of October City winters (rarely below 0°C) without protection
dS/m	Deciseimens per meter: the unit used to measure soil salinity/salt tolerance